

NVIDIA GeForce RTX 4070 Super

Did GPUs just get cheaper?



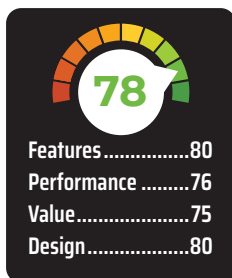
We are talking about the NVIDIA RTX

4070 Super so hold on to your pitchforks until you finish reading this review. Graphics cards have been used for GPGPU software development for a while and now there's a lot of AI development happening at the end-user as well. So we are looking towards making a couple of additions to the way we review graphics cards. In this review, we'll showcase its capabilities in gaming, a bit of content creation and the usual analysis around power consumption.

NVIDIA positions the RTX 4070 Super right between the souped up Ti variant on one side and the base variant on the other with the super basically looking like a slightly younger 4070 ti. Everything from the perspective of the specifications looks really close to the 4070 Ti but it is a lot cheaper. So you can see where this review is leading. Nevertheless, let's see how it actually performs.

SPECIFICATIONS

The RTX 4070 SUPER uses an AD104



GPU but it's not the absolute top-end variant. The RTX 4070 Ti uses a more complete AD104 and has more functional parts available. The NVIDIA RTX 4070 Super is positioned right between the 4070 Ti and the 4070 with 56 Streaming Multiprocessors and 7168

CUDA Cores. Its Shader performance of 36 FLOPs is a significant leap over the RTX 4070, indicating robust shading capabilities crucial for rendering complex scenes. The 224 Tensor Cores with 568 AI TOPS showcase its AI processing prowess, beneficial for AI-enhanced features in games and other AI-driven tasks. This is what folks interested in AI development should be most concerned with, aside from the memory on the board. Moreover, with 56 RT Cores, it offers advanced ray tracing capabilities.

In comparison, the RTX 4080, with 76 Streaming Multiprocessors and 9728 CUDA Cores, overshadows the 4070 SUPER in raw processing power, which is reflected in its higher Shader performance of 49 FLOPs. Its higher Tensor and RT Core count will likely

translate to more efficient AI processing and more realistic ray-traced lighting effects. However, this comes at the cost of higher power consumption at 320 W, compared to the 220 W of the 4070 SUPER.

The RTX 4070 SUPER's boost clock of 2475 MHz is competitive, ensuring that it can handle peak performance demands during intense gaming sessions. That's 495 MHz of headroom and is very similar to what the RTX 4070 has to offer. And like all AD104 GPUs, it features a memory configuration with a 192-bit interface handling 12 GB of GDDR6X memory. This is good enough to fit some small LLMs so this is a good card for folks starting out on AI development courses. As for gaming, there is a nice balance between bandwidth and volume, catering well to modern games that demand both across varying resolutions.

In gaming workloads, these specifications suggest that the RTX 4070 SUPER is a potent card for high-fidelity gaming at high resolutions and frame rates. Its robust AI and ray tracing capabilities will enable it to deliver immersive experiences with advanced visual effects. Moreover, its relatively modest power consumption for its class implies a performance-per-watt proposition that could be attractive to gamers who would rather not see a hefty power bill at the end of the month.

With a core configuration very similar to the RTX 4070 Ti, this does bring into question the need for a 4070 Ti to exist. And a few days before this review, NVIDIA has officially marked the RTX 4070 Ti as End-of-Life (EOL). So you can only get RTX 4070 Ti cards as long as stocks last. Overall, the RTX 4070 SUPER is poised as a high-performance card that offers a sweet spot for gamers and creators who demand high performance without the extremes of the top-tier offerings.

MORE AI FEATURES FOR YOU

We've been running CUDA related benchmarks ever since the first Fermi cards back in 2011 and we've seen how